

Companion Manuscript, Figure 2: Fuzzy Fundamental Theorem of Calculus and Observer-Relative Stokes

Bundle-Valued, Curve-Local FFTC (Part 1, Reduced) and a Companion Stokes Theorem
with Bulk and Boundary Residues

Proofs supporting Figure 2 and Theorems 1–2 of the Hypothesis Surface manuscript (§5,
`main.tex`).

Abstract

This companion manuscript provides the full rigorous proofs for the Fuzzy Fundamental Theorem of Calculus (Part 1, reduced, curve-local) and the Observer-Relative Stokes theorem with bulk and boundary residues, as displayed in Figure 2 of the Hypothesis Surface paper. The bundle-valued formalism developed here (observer-relative symbolic bundle $(E, h_{\mathcal{O}}, \nabla_{\mathcal{O}})$ with kernel-scale $s_{\mathcal{O}}$, connection 1-form $A_{\mathcal{O}}$, and curvature $\kappa_{\mathcal{O}}$) underpins Figure 3’s observer-bounded cohomology and provides the geometric objects referenced by the companion manuscript `fig3_companion_scaling_cohomology.tex`.

0. Notation

Symbols used throughout, grouped by layer. Distinctions from overlapping symbols in the companion Cacophony paper and main Hypothesis Surface paper are noted explicitly.

Ambient geometry. (M, g) : finite-rank smooth Riemannian manifold; $|\mathcal{K}_t|$: open patch of M coordinatized by the simplicial claim complex $\mathcal{K}_t$; $\text{inj}(M, g)$: classical injectivity radius; $\tau_{y \rightarrow x}^{\nabla}$: parallel transport along the g -geodesic from y to x .

Observer data. $s_{\mathcal{O}}$: observer kernel scale (fixed, $< \text{inj}(M, g)$); $K_{\mathcal{O}}^{\text{tan}} : TM \rightarrow TM$: tangent-space smoothing operator; $g_{\mathcal{O}}$: observer-induced fuzzy Riemannian metric (Eq. (1)); $K_{\mathcal{O}} : M \times M \rightarrow \mathbb{R}_{\geq 0}$: section-smoothing scalar kernel (distinct from $K_{\mathcal{O}}^{\text{tan}}$; shared scale $s_{\mathcal{O}}$); $\varepsilon_{\mathcal{O}}$: observer error budget (Def. 6).

Symbolic bundle. $E \rightarrow |\mathcal{K}_t|$: Hermitian complex vector bundle, finite rank; $h_{\mathcal{O}}$: fiber metric; $\nabla_{\mathcal{O}}$: connection, with 1-form $A_{\mathcal{O}} \in \Omega^1(|\mathcal{K}_t|, \text{End}(E))$; $D_{\mathcal{O}} := d + A_{\mathcal{O}}$: symbolic covariant derivative.

Note. The symbol $\hat{\rho}$ is reserved in the companion papers for the *regime index* (Cacophony paper) and *masking-rate companion diagnostic* (main text). We do *not* use $\hat{\rho}$ for $D_{\mathcal{O}}$ here, to avoid collision. $\kappa_{\mathcal{O}} = F_{\mathcal{O}} := dA_{\mathcal{O}} + A_{\mathcal{O}} \wedge A_{\mathcal{O}} \in \Omega^2(|\mathcal{K}_t|, \text{End}(E))$: curvature 2-form. **Distinct** from κ_t in main-text Prop. 2 (condition number of the claim-interaction matrix $\tilde{\Gamma}_t$).

FFTC objects. $f \in \Gamma(E)$: epistemic field (section); F_{γ} : reduced observer-bounded integral along γ (Def. 5; frame-relative); $\eta_{\mathcal{O}}$: kernel-smoothing residue (Def. 6); $\rho_{\mathcal{O}}$: FFTC residue (Cor. 1), a section along γ ; **distinct** from the 2-form $\kappa_{\mathcal{O}}$; $\mathcal{I}_{\mathcal{O}}, \mathcal{H}_{\mathcal{O}}$: bulk interaction and boundary holonomy residues (Thm. 1).

Anti-masking bridge. μ : masking rate (Invariant 2 of the main text), a text-level diagnostic. The map from μ to bundle-level bounds on $\rho_{\mathcal{O}}$ is declarative in Cor. 2 and scoped to adjacent work.

1. Setup

Let $\mathcal{K}_t$ be the simplicial claim complex at time t : vertices are claims C_i carrying fibers $\phi_i = (\sigma_i, r_i, \mathcal{W}_i)$ with resolution $r_i \in (0, \infty]$. Fix an observer kernel scale $s_{\mathcal{O}} > 0$ governing the smoothing radius below.

Ambient smooth substrate. We do *not* reconstruct smooth structure from $\mathcal{K}_t$. Instead, we identify $|\mathcal{K}_t|$ with (an open patch of) a smooth Riemannian manifold (M, g) of finite rank n ; the simplicial skeleton $\mathcal{K}_t$ serves as a combinatorial coordinatization of M , and all geometric operations are carried out on (M, g) . Piecewise-smooth paths $\gamma : [a, b] \rightarrow M$ are well-defined in the usual sense; we require only piecewise smoothness, not global smoothness of γ . This inherits smooth structure rather than producing it from $\mathcal{K}_t$, and sidesteps the CW-vs.-smooth passage entirely.

The observer-induced metric $g_{\mathcal{O}}$. Let $K_{\mathcal{O}}^{\tan} : TM \rightarrow TM$ be a smoothing operator on the tangent bundle with characteristic scale $s_{\mathcal{O}}$, fiber-preserving ($K_{\mathcal{O}}^{\tan}(T_p M) \subseteq T_p M$) and self-adjoint with respect to g . The *observer-induced metric* is

$$g_{\mathcal{O}}(p)(v, w) := \langle K_{\mathcal{O}}^{\tan} v, K_{\mathcal{O}}^{\tan} w \rangle_{g(p)}, \quad v, w \in T_p M. \quad (1)$$

Basic properties are elementary; we record them for later use.

Lemma 1 (Properties of $g_{\mathcal{O}}$). *The bilinear form $g_{\mathcal{O}}$ defined by Eq. (1) satisfies:*

- (i) $g_{\mathcal{O}}(v, w) = g_{\mathcal{O}}(w, v)$ (*symmetric*).
- (ii) $g_{\mathcal{O}}(v, v) \geq 0$, with equality iff $K_{\mathcal{O}}^{\tan} v = 0$ (*positive-semidefinite*; $\ker K_{\mathcal{O}}^{\tan} \subseteq T_p M$ is the space of tangent directions invisible to $\mathcal{O}$).

In particular, $g_{\mathcal{O}}$ is a pseudo-metric in the sense that $g_{\mathcal{O}}(v, v) = 0$ does not force $v = 0$: sub- $s_{\mathcal{O}}$ tangent directions are $\mathcal{O}$ -indistinguishable, which is the geometric meaning of bounded observation.

Proof. (i) $\langle K_{\mathcal{O}}^{\tan} v, K_{\mathcal{O}}^{\tan} w \rangle_g = \langle v, (K_{\mathcal{O}}^{\tan})^2 w \rangle_g = \langle (K_{\mathcal{O}}^{\tan})^2 v, w \rangle_g = \langle K_{\mathcal{O}}^{\tan} w, K_{\mathcal{O}}^{\tan} v \rangle_g$ by the self-adjointness of $K_{\mathcal{O}}^{\tan}$. (ii) $g_{\mathcal{O}}(v, v) = \|K_{\mathcal{O}}^{\tan} v\|_g^2 \geq 0$, with equality iff $K_{\mathcal{O}}^{\tan} v = 0$. $\square$

Injectivity radius. We assume $s_{\mathcal{O}} < \text{inj}(M, g)$. This ensures the exponential map and parallel transport along geodesics are well-defined on $s_{\mathcal{O}}$ -balls, which is used in Def. 3 (fiberwise convolution) and Thm. 1 (closing-arc η).

Remark 1 (Scope of the passage from $\mathcal{K}_t$ to M). *The discrete exterior calculus of Desbrun–Hirani (2003) is a heavyweight alternative that constructs calculus intrinsically on simplicial complexes; the smooth-substrate route adopted here is lighter and suffices for the Hypothesis Surface paper’s purposes.*

2. The Observer-Relative Symbolic Bundle

We work in a bundle-valued formalism, which matches the gauge-theoretic framing of the main paper’s companion manuscript and avoids the scalar/non-abelian mismatch that earlier drafts exhibited.

Definition 1 (Observer-Relative Symbolic Bundle). *Let $(E, h_{\mathcal{O}}) \rightarrow |\mathcal{K}_t|$ be a Hermitian complex vector bundle of finite rank with smoothly varying fiber metric $h_{\mathcal{O}}$. An observer-relative symbolic bundle is a triple $(E, h_{\mathcal{O}}, \nabla_{\mathcal{O}})$ where $\nabla_{\mathcal{O}}$ is a metric-compatible affine connection on E with connection 1-form $A_{\mathcal{O}} \in \Omega^1(|\mathcal{K}_t|, \text{End}(E))$, assumed $\mathcal{O}$ -bounded: $\|A_{\mathcal{O}}\|_{g_{\mathcal{O}} \otimes h_{\mathcal{O}}, \infty} < \infty$, where*

$$\|A_{\mathcal{O}}\|_{g_{\mathcal{O}} \otimes h_{\mathcal{O}}, \infty} := \sup_{x \in |\mathcal{K}_t|} \sup_{\substack{v \in T_x |\mathcal{K}_t| \\ \|v\|_{g_{\mathcal{O}}} = 1}} \|A_{\mathcal{O}}(x)[v]\|_{\text{End}(E_x), h_{\mathcal{O}}}, \quad (2)$$

and $\|\cdot\|_{\text{End}(E_x), h_{\mathcal{O}}}$ is the operator norm on $\text{End}(E_x)$ induced by $h_{\mathcal{O}}$ (i.e. $\|T\| := \sup_{\|w\|_{h_{\mathcal{O}}}=1} \|Tw\|_{h_{\mathcal{O}}}$). The symbolic curvature is

$$\kappa_{\mathcal{O}} = F_{\mathcal{O}} := dA_{\mathcal{O}} + A_{\mathcal{O}} \wedge A_{\mathcal{O}} \in \Omega^2(|\mathcal{K}_t|, \text{End}(E)),$$

well-defined because $A_{\mathcal{O}}$ is $\text{End}(E)$ -valued (the wedge uses composition in $\text{End}(E)$). Epistemic fields are sections $f \in \Gamma(E)$.

Remark 2 (The sup in Eq. (2) is over $\mathcal{O}$ -visible directions). *Because $g_{\mathcal{O}}$ is degenerate on $\ker K_{\mathcal{O}}^{\text{tan}}$ (Lem. 1(ii)), the inner sup in Eq. (2) effectively ranges only over $\mathcal{O}$ -visible unit tangent directions: $A_{\mathcal{O}}$'s action on $\ker K_{\mathcal{O}}^{\text{tan}}$ is unconstrained by $\|A_{\mathcal{O}}\|_{g_{\mathcal{O}} \otimes h_{\mathcal{O}}, \infty} < \infty$. This is harmless for curve-local results because the FFTC (Thm. 2) requires $\|\dot{\gamma}\|_{g_{\mathcal{O}}} > 0$ on smooth segments, which forbids $\dot{\gamma} \in \ker K_{\mathcal{O}}^{\text{tan}}$; the integrand in Thm. 1 similarly couples $A_{\mathcal{O}}$ only to $\mathcal{O}$ -visible tangents.*

Definition 2 (Observer-Relative Symbolic Space on $\mathcal{K}_t$). *The observer-relative symbolic space over $\mathcal{K}_t$ is the tuple $(|\mathcal{K}_t|, g_{\mathcal{O}}, E, h_{\mathcal{O}}, \nabla_{\mathcal{O}}, \kappa_{\mathcal{O}})$ where $g_{\mathcal{O}}$ is the fuzzy Riemannian metric on $|\mathcal{K}_t|$ (Eq. (1), with kernel scale $s_{\mathcal{O}}$: distances below $s_{\mathcal{O}}$ are indistinguishable under $g_{\mathcal{O}}$), and $(E, h_{\mathcal{O}}, \nabla_{\mathcal{O}})$ is the symbolic bundle of Def. 1.*

Definition 3 (Observer Convolution Kernel (on E -sections)). *Distinct from the tangent-space kernel $K_{\mathcal{O}}^{\text{tan}}$ of Eq. (1): let $K_{\mathcal{O}} : M \times M \rightarrow \mathbb{R}_{\geq 0}$ be a smooth, symmetric, normalized ($\int K_{\mathcal{O}}(x, y) dy_g = 1$, where dy_g is the volume form of the ambient g) scalar kernel with support of radius $s_{\mathcal{O}}$ in the g -metric. For $f \in \Gamma(E)$, convolution acts fiberwise via parallel transport along $\nabla_{\mathcal{O}}$:*

$$(K_{\mathcal{O}} * f)(x) := \int K_{\mathcal{O}}(x, y) \tau_{y \rightarrow x}^{\nabla_{\mathcal{O}}} [f(y)] dy_g,$$

where $\tau_{y \rightarrow x}^{\nabla_{\mathcal{O}}} : E_y \rightarrow E_x$ is parallel transport along the unique g -geodesic from y to x inside the $s_{\mathcal{O}}$ -ball (well-defined by $s_{\mathcal{O}} < \text{inj}(M, g)$). Both kernels share the scale $s_{\mathcal{O}}$; $K_{\mathcal{O}}^{\text{tan}}$ governs the metric $g_{\mathcal{O}}$, while $K_{\mathcal{O}}$ governs section smoothing. As $s_{\mathcal{O}} \rightarrow 0$, $K_{\mathcal{O}} \rightarrow \delta$ and $\tau^{\nabla_{\mathcal{O}}} \rightarrow \text{id}$ on the (vanishing) support, recovering pointwise evaluation of f .

Remark 3 (Choice of ambient measure). *The volume form dy_g and the kernel's support radius are taken with respect to the ambient metric g , not $g_{\mathcal{O}}$: $g_{\mathcal{O}}$ is degenerate (Lem. 1) and carries no natural volume form. This is a deliberate asymmetry of the observer-relative framework. The observer sets the scale $s_{\mathcal{O}}$; the measure against which $K_{\mathcal{O}}$ is normalized is the classical one. The observer's bounded resolution manifests in the support size of $K_{\mathcal{O}}$ (at scale $s_{\mathcal{O}}$) and in the degeneracy of $g_{\mathcal{O}}$, not in the base measure. No result below requires an observer-relative volume form.*

Definition 4 (Symbolic Covariant Derivative). *The symbolic covariant derivative is the exterior covariant derivative induced by $\nabla_{\mathcal{O}}$:*

$$D_{\mathcal{O}} = d^{\nabla_{\mathcal{O}}} : \Gamma(E) \rightarrow \Omega^1(|\mathcal{K}_t|, E), \quad D_{\mathcal{O}} f = df + A_{\mathcal{O}} \cdot f,$$

where $A_{\mathcal{O}} \cdot f$ is the fibrewise action of $A_{\mathcal{O}} \in \text{End}(E)$ on $f \in E$. In the classical differentiation-integration duality, $D_{\mathcal{O}}$ is the fuzzy-symbolic analogue of d/dx : it resolves an epistemic state into conflict components.

Definition 5 (Observer-Bounded Integration along a Curve, Reduced). This is a frame-relative construction. Fix a piecewise-smooth curve $\gamma : [a, b] \rightarrow |\mathcal{K}_t|$ and a smooth local trivialization $\Phi : E|_U \cong U \times \mathbb{C}^k$ over a tubular neighborhood $U \supset \gamma([a, b])$. In Φ , the kernel-smoothed field $(K_{\mathcal{O}} * f)(x) \in E_x$ has components $(K_{\mathcal{O}} * f)^\alpha(x) \in \mathbb{C}$, $\alpha = 1, \dots, k$. The reduced observer-bounded integral along γ in Φ is the map $[a, b] \rightarrow E$ with $F_\gamma(s) \in E_{\gamma(s)}$ given by componentwise integration in Φ :

$$F_\gamma^\alpha(s) := \int_a^s (K_{\mathcal{O}} * f)^\alpha(\gamma(u)) \|\dot{\gamma}(u)\|_{g_{\mathcal{O}}} du, \quad F_\gamma(s) := F_\gamma^\alpha(s) e_\alpha(\gamma(s)),$$

where $\{e_\alpha\}$ is the frame associated to Φ . Throughout Sections 3–5, the trivialization Φ on U is fixed, and all results below are asserted relative to Φ ; frame-dependence is tracked explicitly in Rmk. 5.

Remark 4 (Reduced operator vs. full symbolic-integration operator). Def. 5 proves a reduced FFTC. A fuller operator, following the companion manuscript, augments this with a memory/distortion term $E_{\text{acc}}(\gamma, f)$ capturing tangent blur and homotopy-class residues. The reduced version is the load-bearing object for the main paper’s integrability claim ($\mu = 0$ at the kernel/metric level); the E_{acc} contribution is a separate layer whose local piece can be absorbed into $\rho_{\mathcal{O}}(f, \gamma)$ under $\mathcal{O}$ -regularity and whose topological piece is Part 2 territory (not established here). We flag this scope boundary explicitly rather than silently redefining the fuller operator.

Remark 5 (Frame-dependence of F_γ ; compatibility under frame changes). F_γ is a frame-relative construction: different smooth trivializations of $E|_U$ yield different sections along γ . Under a smooth frame change $\Phi \rightarrow \Psi$ with transition $g : U \rightarrow GL(k, \mathbb{C})$ (so $e_\alpha \mapsto g_\alpha^\beta(x) e_\beta$), the components of $(K_{\mathcal{O}} * f)$ transform as $(K_{\mathcal{O}} * f)^{\tilde{\beta}}(x) = (g^{-1}(x))_\alpha^\beta (K_{\mathcal{O}} * f)^\alpha(x)$. Re-computing Def. 5 in Ψ yields

$$\tilde{F}_\gamma^\beta(s) = \int_a^s (g^{-1}(\gamma(u)))_\alpha^\beta (K_{\mathcal{O}} * f)^\alpha(\gamma(u)) \|\dot{\gamma}\|_{g_{\mathcal{O}}} du,$$

which in general differs from $(g^{-1}(\gamma(s)))_\alpha^\beta F_\gamma^\alpha(s)$ whenever g varies along γ : the pointwise pullback does not commute with the u -integral. This is the familiar path-ordering obstruction. Two relevant regimes:

- (a) Rigid frame changes ($dg \equiv 0$ on γ , e.g., g constant on γ): g commutes with the integral, so $\tilde{F}_\gamma = g^{-1} F_\gamma$ pointwise, and Eq. (8) transforms covariantly.
- (b) Position-dependent frame changes ($dg \not\equiv 0$ on γ): Eq. (8) is a family of identities indexed by Φ ; in each chosen Φ , the identity holds (same proof), with the connection-coupling term $A_{\mathcal{O}}[T_\gamma] \cdot F_\gamma$ absorbing the path-ordering dependence. See Step 5 of the proof of Thm. 2.

The norm bound Cor. 1 is frame-independent up to the bounded frame-comparison constant C_{frame} of Lem. 2; downstream users (Cor. 2) rely only on this bound, not on a strict geometric invariance of F_γ itself.

Lemma 2 (Frame-comparison constant). Let Φ, Ψ be two smooth trivializations of E over an open $U \subset |\mathcal{K}_t|$, with transition $g : U \rightarrow GL(k, \mathbb{C})$. On any compact $K \subset U$ there exists $C_{\text{frame}} = C_{\text{frame}}(K, \Phi, \Psi, h_{\mathcal{O}}) \in (0, \infty)$ such that for every $v \in E_x$ with $x \in K$,

$$C_{\text{frame}}^{-1} \|v\|_{\Phi, 2} \leq \|v\|_{h_{\mathcal{O}}} \leq C_{\text{frame}} \|v\|_{\Psi, 2},$$

where $\|\cdot\|_{\Phi, 2}$ denotes the Euclidean norm on components in the frame Φ , and analogously for Ψ . Consequently, norm bounds obtained via componentwise estimates in Φ imply norm bounds in $h_{\mathcal{O}}$ up to the factor C_{frame} , and any two such estimates for a given section differ by a factor $\leq C_{\text{frame}}^2$.

Proof. $h_{\mathcal{O}}$ is smooth and positive-definite on fibers; the frame Φ induces a smooth fiber-wise Hermitian Gramian matrix $H_{\Phi}(x)$ relating $\|\cdot\|_{\Phi,2}$ to $\|\cdot\|_{h_{\mathcal{O}}}$. H_{Φ} is continuous on the compact K , so it is uniformly bounded above and below; take C_{frame} as the square root of the ratio of its largest to smallest eigenvalues over K . $\square$

Definition 6 ($\mathcal{O}$ -Regularity). *A section $f \in \Gamma(E)$ is $\mathcal{O}$ -regular if there exists $\varepsilon_{\mathcal{O}} < \infty$ such that*

$$\|(K_{\mathcal{O}} * f)(x) - f(x)\|_{h_{\mathcal{O}}} \leq \varepsilon_{\mathcal{O}} \quad \forall x \in |\mathcal{K}_t|.$$

The scalar $\varepsilon_{\mathcal{O}}$ is the observer's sub-threshold error budget: kernel smoothing perturbs f pointwise in the fiber norm by at most $\varepsilon_{\mathcal{O}}$.

Definition 7 ($\mathcal{O}$ -Closed Loop). *A piecewise-smooth path $\gamma : [a, b] \rightarrow |\mathcal{K}_t|$ is $\mathcal{O}$ -closed if $\|\gamma(a) - \gamma(b)\|_{g_{\mathcal{O}}} \leq s_{\mathcal{O}}$. Equivalently, $\gamma(a)$ and $\gamma(b)$ project to the same equivalence class under the $s_{\mathcal{O}}$ -quotient of $|\mathcal{K}_t|$: the endpoints are indistinguishable at the observer's resolution.*

Remark 6 ($\mathcal{O}$ -closed vs. classically closed). *A classically closed loop ($\gamma(a) = \gamma(b)$) is $\mathcal{O}$ -closed, but not conversely. Exact closure is an unbounded idealization; $\mathcal{O}$ -closure is the natural form for a bounded observer.*

3. Observer-Relative Stokes (Companion Theorem)

The theorem below is a second, independent result on the same observer-relative symbolic bundle. It is *not* used in the proof of Theorem 2: the FFTC is a pointwise along- γ statement, while the result below is loop-integrated. The two theorems share the connection $\nabla_{\mathcal{O}}$ but extract different scalars from it. The Stokes theorem is the load-bearing result for the main paper's *loop-level* integrability arguments.

Remark 7 (Bundle triviality over Σ). *In the Stokes proof below, Σ is a simply connected 2-manifold with nonempty boundary. Such a surface deformation retracts to its 1-skeleton under a handle decomposition, and the 1-skeleton is a simply connected graph, i.e., a tree, hence contractible. Therefore Σ is contractible, and every smooth vector bundle over Σ is trivial. We may choose a global smooth trivialization of $E|_{\Sigma}$, and the componentwise Stokes step below is justified without further obstruction.*

Theorem 1 (Observer-Relative Stokes with Bulk and Boundary Residues). *Let $\Sigma \subset |\mathcal{K}_t|$ be a simply connected surface whose boundary $\partial\Sigma$ is an $\mathcal{O}$ -closed loop. Fix a global smooth trivialization of $E|_{\Sigma}$ (exists by Rmk. 7). For any section $f \in \Gamma(E)$,*

$$\oint_{\partial\Sigma}^{\mathcal{O}} D_{\mathcal{O}} f = \iint_{\Sigma} \kappa_{\mathcal{O}} \cdot f + \mathcal{I}_{\mathcal{O}}(f, \Sigma) + \mathcal{H}_{\mathcal{O}}(\partial\Sigma, f), \quad (3)$$

where $\kappa_{\mathcal{O}} \cdot f$ denotes the fiberwise action of the curvature 2-form on the section f , the bulk inter-action residue

$$\mathcal{I}_{\mathcal{O}}(f, \Sigma) := - \iint_{\Sigma} A_{\mathcal{O}} \wedge D_{\mathcal{O}} f \quad (4)$$

is the coupling of the connection to its own covariant variation, and $\mathcal{H}_{\mathcal{O}}$ is a boundary holonomy term defined in step (a) of the proof. Both residues are $\mathcal{O}$ -bounded:

$$\|\mathcal{H}_{\mathcal{O}}(\partial\Sigma, f)\|_{h_{\mathcal{O}}} \leq s_{\mathcal{O}} \cdot (L_f + \|A_{\mathcal{O}}\|_{g_{\mathcal{O}} \otimes h_{\mathcal{O}}, \infty} \cdot \|f\|_{\Sigma, \infty}), \quad (5)$$

$$\|\mathcal{I}_{\mathcal{O}}(f, \Sigma)\|_{h_{\mathcal{O}}} \leq \text{Area}(\Sigma) \cdot \|A_{\mathcal{O}}\|_{g_{\mathcal{O}} \otimes h_{\mathcal{O}}, \infty} \cdot (L_f + \|A_{\mathcal{O}}\|_{g_{\mathcal{O}} \otimes h_{\mathcal{O}}, \infty} \cdot \|f\|_{\Sigma, \infty}), \quad (6)$$

with L_f the piecewise-Lipschitz constant of f at scale $s_{\mathcal{O}}$ under (H3). Both residues vanish in the trivial-connection limit $A_{\mathcal{O}} \equiv 0$; $\mathcal{H}_{\mathcal{O}}$ additionally vanishes in the closing-arc limit $s_{\mathcal{O}} \rightarrow 0$, and $\mathcal{I}_{\mathcal{O}}$ vanishes whenever $\|A_{\mathcal{O}}\|_{g_{\mathcal{O}} \otimes h_{\mathcal{O}}, \infty} \rightarrow 0$.

Proof. (a) *Reduction.* Let $p := \gamma(a)$ and $q := \gamma(b)$ be the endpoints of $\partial\Sigma = \gamma$. By Def. 7, $\|p - q\|_{g_{\mathcal{O}}} \leq s_{\mathcal{O}}$. Pick a smooth path $\eta : [0, 1] \rightarrow |\mathcal{K}_t|$ from q to p with arc-length $\leq s_{\mathcal{O}}$ (exists by geodesic connectivity inside the $s_{\mathcal{O}}$ -ball around p , using $s_{\mathcal{O}} < \text{inj}(M, g)$). The augmented loop $\tilde{\gamma} := \gamma \star \eta$ is classically closed. Applying classical Stokes to $D_{\mathcal{O}}f$ treated componentwise as an ordinary bundle-valued 1-form in the trivialization of Rmk. 7 yields the corrected covariant identity,¹

$$\oint_{\tilde{\gamma}} D_{\mathcal{O}}f = \iint_{\Sigma} \kappa_{\mathcal{O}} \cdot f + \mathcal{I}_{\mathcal{O}}(f, \Sigma). \quad (7)$$

By additivity over concatenation, $\oint_{\tilde{\gamma}} D_{\mathcal{O}}f = \oint_{\gamma} D_{\mathcal{O}}f + \int_{\eta} D_{\mathcal{O}}f$. Define $\mathcal{H}_{\mathcal{O}}(\partial\Sigma, f) := -\int_{\eta} D_{\mathcal{O}}f$; rearranging gives Eq. (3).

(b) *Bound on $\mathcal{H}_{\mathcal{O}}$.* Expanding $D_{\mathcal{O}}f = df + A_{\mathcal{O}} \cdot f$ and integrating along η :

$$\int_{\eta} D_{\mathcal{O}}f = \int_{\eta} df + \int_{\eta} A_{\mathcal{O}} \cdot f.$$

The df -term telescopes: $\int_{\eta} df = f(p) - f(q)$, which under (H3) (piecewise Lipschitz at scale $s_{\mathcal{O}}$) is bounded in fiber norm by $L_f \cdot \|p - q\|_{g_{\mathcal{O}}} \leq L_f \cdot s_{\mathcal{O}}$.² The $A_{\mathcal{O}}$ -term is bounded by the arc-length of η ($\leq s_{\mathcal{O}}$) times the sup-norm of the integrand ($\leq \|A_{\mathcal{O}}\|_{g_{\mathcal{O}} \otimes h_{\mathcal{O}}, \infty} \cdot \|f\|_{\Sigma, \infty}$). Combining yields Eq. (5).

(c) *Bound on $\mathcal{I}_{\mathcal{O}}$.* Pointwise in the global frame, $\|A_{\mathcal{O}} \wedge D_{\mathcal{O}}f\|_{h_{\mathcal{O}}} \leq \|A_{\mathcal{O}}\|_{g_{\mathcal{O}} \otimes h_{\mathcal{O}}, \infty} \cdot \|D_{\mathcal{O}}f\|_{h_{\mathcal{O}}} \leq \|A_{\mathcal{O}}\|_{g_{\mathcal{O}} \otimes h_{\mathcal{O}}, \infty} \cdot (L_f + \|A_{\mathcal{O}}\|_{g_{\mathcal{O}} \otimes h_{\mathcal{O}}, \infty} \cdot \|f\|_{\Sigma, \infty})$, using $D_{\mathcal{O}}f = df + A_{\mathcal{O}}f$ and (H3). Integrating over Σ yields Eq. (6). $\square$

Remark 8 (Dependence of $\mathcal{H}_{\mathcal{O}}$ on the interpolation path η). $\mathcal{H}_{\mathcal{O}}$ depends on the choice of interpolation path η from q to p (Def. 7 controls the endpoint $g_{\mathcal{O}}$ -distance, not the path). The dependence is controlled directly, without recursion: for any two admissible choices η, η' (each with arc-length $\leq s_{\mathcal{O}}$), additivity of line integrals gives

$$\mathcal{H}_{\mathcal{O}}(\partial\Sigma, f; \eta) - \mathcal{H}_{\mathcal{O}}(\partial\Sigma, f; \eta') = -\int_{\eta} D_{\mathcal{O}}f + \int_{\eta'} D_{\mathcal{O}}f = \int_{\eta' \star \bar{\eta}} D_{\mathcal{O}}f,$$

and the concatenated loop $\eta' \star \bar{\eta}$ has arc-length $\leq 2s_{\mathcal{O}}$. Applying the Step (b) estimate to this loop,

$$\|\mathcal{H}_{\mathcal{O}}(\partial\Sigma, f; \eta) - \mathcal{H}_{\mathcal{O}}(\partial\Sigma, f; \eta')\|_{h_{\mathcal{O}}} \leq 2s_{\mathcal{O}} \cdot (L_f + \|A_{\mathcal{O}}\|_{g_{\mathcal{O}} \otimes h_{\mathcal{O}}, \infty} \cdot \|f\|_{\Sigma, \infty}).$$

Hence the norm bound (5) is stable across choices of η , up to the factor 2; no recursion is needed.

¹Since $D_{\mathcal{O}} = d + A_{\mathcal{O}}$ acts on sections in a local frame, classical Stokes applied to the ordinary 1-form $D_{\mathcal{O}}f = df + A_{\mathcal{O}}f$ gives $\oint_{\tilde{\gamma}} D_{\mathcal{O}}f = \iint_{\Sigma} d(df + A_{\mathcal{O}}f) = \iint_{\Sigma} (dA_{\mathcal{O}} \wedge f - A_{\mathcal{O}} \wedge df) = \iint_{\Sigma} (dA_{\mathcal{O}} + A_{\mathcal{O}} \wedge A_{\mathcal{O}}) \cdot f - \iint_{\Sigma} A_{\mathcal{O}} \wedge D_{\mathcal{O}}f = \iint_{\Sigma} \kappa_{\mathcal{O}} \cdot f - \iint_{\Sigma} A_{\mathcal{O}} \wedge D_{\mathcal{O}}f$, using $d(df) = 0$ and $df = D_{\mathcal{O}}f - A_{\mathcal{O}}f$ (hence $-A_{\mathcal{O}} \wedge df = -A_{\mathcal{O}} \wedge D_{\mathcal{O}}f + A_{\mathcal{O}} \wedge A_{\mathcal{O}} \cdot f$) and the curvature identity $\kappa_{\mathcal{O}} = dA_{\mathcal{O}} + A_{\mathcal{O}} \wedge A_{\mathcal{O}}$. The extra surface term is the interaction residue $\mathcal{I}_{\mathcal{O}}$; the naive reading $\oint D_{\mathcal{O}}f = \iint (D_{\mathcal{O}})^2 f$ is invalid because $D_{\mathcal{O}}$ is a covariant, not exterior, derivative, so Stokes does not apply to it directly. For the covariant exterior derivative and the curvature identity $(d^{\nabla})^2 = F^{\nabla}$, see Kobayashi and Nomizu, *Foundations of Differential Geometry, Vol. II*, Ch. III; for classical Stokes applied componentwise in a trivialization, Bott and Tu, *Differential Forms in Algebraic Topology*, Ch. I. These textbook references are self-contained and do not invoke the companion manuscript.

²We do not derive Lipschitz-at-scale- $s_{\mathcal{O}}$ from $\mathcal{O}$ -regularity. The two hypotheses are independent: $\mathcal{O}$ -regularity controls $\|K_{\mathcal{O}} * f - f\|$ (a pointwise kernel-smoothing error), not $\|\nabla f\|$ (a pointwise differentiability modulus). (H3) is posted separately; its compatibility with pipeline-emitted fields is discussed in Rmk. 12, not within the hypothesis itself.

4. Main Theorem (FFTC Part 1, Curve-Local, Reduced)

Theorem 2 (FFTC Part 1, curve-local bundle form). *Let $(E, h_{\mathcal{O}}, \nabla_{\mathcal{O}})$ be an observer-relative symbolic bundle over $|\mathcal{K}_t|$ with $s_{\mathcal{O}} < \text{inj}(M, g)$. Let $f \in \Gamma(E)$ be $\mathcal{O}$ -regular (Def. 6) and piecewise Lipschitz at scale $s_{\mathcal{O}}$ (H3). Let $\gamma : [a, b] \rightarrow |\mathcal{K}_t|$ be piecewise smooth, regular ($\|\dot{\gamma}(s)\|_{g_{\mathcal{O}}} > 0$ where smooth, so that $\dot{\gamma} \notin \ker K_{\mathcal{O}}^{\text{tan}}$) and injective on $[a, b]$, with unit $g_{\mathcal{O}}$ -tangent $T_{\gamma}(s) := \dot{\gamma}(s)/\|\dot{\gamma}(s)\|_{g_{\mathcal{O}}}$. Fix a smooth local trivialization Φ of E on a tubular neighborhood of $\gamma([a, b])$, and let F_{γ} be the reduced observer-bounded integral of Def. 5 in Φ . Then for each $s \in [a, b]$ at which γ is smooth,*

$$(D_{\mathcal{O}}F_{\gamma})(\gamma(s))[T_{\gamma}(s)] = f(\gamma(s)) + \eta_{\mathcal{O}}(\gamma(s)) + A_{\mathcal{O}}(\gamma(s))[T_{\gamma}(s)] \cdot F_{\gamma}(s), \quad (8)$$

where $\eta_{\mathcal{O}}(x) := (K_{\mathcal{O}} * f)(x) - f(x)$ is the kernel-smoothing residue, with $\|\eta_{\mathcal{O}}(x)\|_{h_{\mathcal{O}}} \leq \varepsilon_{\mathcal{O}}$ for all x (Def. 6). The identity (8) is asserted in the chosen trivialization Φ ; under frame changes it behaves as described in Step 5 of the proof and Rmk. 5. The norm bound on the residue (Cor. 1) is frame-independent up to C_{frame} (Lem. 2).

Proof. We compute $(D_{\mathcal{O}}F_{\gamma})(\gamma(s))[T_{\gamma}(s)]$ in the fixed local trivialization Φ of Def. 5, then invoke $\mathcal{O}$ -regularity.

Step 1 (derivative of the integral, in the frame). In $\Phi : E|_U \cong U \times \mathbb{C}^k$, write $F_{\gamma}(s) = F_{\gamma}^{\alpha}(s) e_{\alpha}(\gamma(s))$ and $(K_{\mathcal{O}} * f)(x) = (K_{\mathcal{O}} * f)^{\alpha}(x) e_{\alpha}(x)$. By Def. 5, each component F_{γ}^{α} is an ordinary scalar integral; the classical fundamental theorem of calculus gives

$$\left. \frac{dF_{\gamma}^{\alpha}}{ds} \right|_s = (K_{\mathcal{O}} * f)^{\alpha}(\gamma(s)) \cdot \|\dot{\gamma}(s)\|_{g_{\mathcal{O}}}, \quad \alpha = 1, \dots, k.$$

Step 2 (expanding $D_{\mathcal{O}}$ in the frame). Def. 4 gives $D_{\mathcal{O}}F_{\gamma} = dF_{\gamma} + A_{\mathcal{O}} \cdot F_{\gamma}$. Contracting with the unit tangent $T_{\gamma}(s)$:

$$(D_{\mathcal{O}}F_{\gamma})(\gamma(s))[T_{\gamma}(s)] = (dF_{\gamma})(\gamma(s))[T_{\gamma}(s)] + A_{\mathcal{O}}(\gamma(s))[T_{\gamma}(s)] \cdot F_{\gamma}(s).$$

The first term, componentwise, is

$$(dF_{\gamma}^{\alpha})(\gamma(s))[T_{\gamma}(s)] = \left. \frac{dF_{\gamma}^{\alpha}}{ds} \right|_s \cdot \frac{1}{\|\dot{\gamma}(s)\|_{g_{\mathcal{O}}}} = (K_{\mathcal{O}} * f)^{\alpha}(\gamma(s)) \quad (\text{by Step 1}).$$

Reassembling components: $(dF_{\gamma})(\gamma(s))[T_{\gamma}(s)] = (K_{\mathcal{O}} * f)(\gamma(s))$.

Step 3 ($\mathcal{O}$ -regularity closes the kernel error). By Def. 6, $(K_{\mathcal{O}} * f)(\gamma(s)) = f(\gamma(s)) + \eta_{\mathcal{O}}(\gamma(s))$ with $\|\eta_{\mathcal{O}}\|_{h_{\mathcal{O}}} \leq \varepsilon_{\mathcal{O}}$. Substituting into Step 2 yields Eq. (8).

Step 4 (bounds on the residue). The kernel-smoothing residue is bounded in fiber norm by $\varepsilon_{\mathcal{O}}$ (Def. 6). For the connection-coupling term, by Eq. (2) and the operator-norm definition, $\|A_{\mathcal{O}}(x)[T_{\gamma}(s)] \cdot F_{\gamma}(s)\|_{h_{\mathcal{O}}} \leq \|A_{\mathcal{O}}\|_{g_{\mathcal{O}} \otimes h_{\mathcal{O}}, \infty} \cdot \|F_{\gamma}(s)\|_{h_{\mathcal{O}}}$. We bound $\|F_{\gamma}(s)\|_{h_{\mathcal{O}}}$ explicitly: by Def. 5 and the triangle inequality (componentwise in Φ , then converted to the fiber norm $h_{\mathcal{O}}$ via C_{frame} on the compact $\gamma([a, b])$, per Lem. 2),

$$\|F_{\gamma}(s)\|_{h_{\mathcal{O}}} \leq C_{\text{frame}} \int_a^s \|(K_{\mathcal{O}} * f)(\gamma(u))\|_{h_{\mathcal{O}}} \|\dot{\gamma}(u)\|_{g_{\mathcal{O}}} du \leq C_{\text{frame}} \cdot (\|f\|_{\infty} + \varepsilon_{\mathcal{O}}) \cdot L(\gamma|_{[a, s]}),$$

where $L(\gamma|_{[a, s]}) := \int_a^s \|\dot{\gamma}\|_{g_{\mathcal{O}}} du$ is $g_{\mathcal{O}}$ -arc-length and $\|f\|_{\infty} := \sup_{x \in \gamma([a, b])} \|f(x)\|_{h_{\mathcal{O}}}$ is finite by continuity of f on a compact image. Hence $\|F_{\gamma}\|_{h_{\mathcal{O}}}$ is uniformly bounded on $[a, b]$, and the connection-coupling term is $\mathcal{O}$ -bounded.

Step 5 (compatibility under frame changes). Under a *rigid* frame change $e_\alpha \rightarrow g_\alpha^\beta e_\beta$ with $g \in GL(k, \mathbb{C})$ constant on U (equivalently $dg \equiv 0$), g commutes with the scalar integrals of Def. 5, so $F_\gamma \mapsto gF_\gamma$ pointwise; meanwhile $A_\mathcal{O} \mapsto gA_\mathcal{O}g^{-1}$ (the $g^{-1}dg$ inhomogeneous term vanishes), $D_\mathcal{O}F_\gamma \mapsto gD_\mathcal{O}F_\gamma$, and both sides of Eq. (8) transform by g .

Under a *smooth position-dependent* frame change $g : U \rightarrow GL(k, \mathbb{C})$ with $dg \not\equiv 0$ on γ , F_γ does *not* transform as $g \cdot F_\gamma$: Rmk. 5 shows the new-frame integral $\tilde{F}_\gamma^\beta$ differs from $(g^{-1})_\alpha^\beta F_\gamma^\alpha$ by a path-ordering correction. However, Eq. (8) still holds in the new frame, because Steps 1–4 apply in any chosen trivialization: re-running them in Ψ (with transition g) produces the identity $(\tilde{D}_\mathcal{O}\tilde{F}_\gamma)[T_\gamma] = f + \eta_\mathcal{O} + \tilde{A}_\mathcal{O}[T_\gamma] \cdot \tilde{F}_\gamma$ where $\tilde{A}_\mathcal{O} = g^{-1}A_\mathcal{O}g + g^{-1}dg$. The connection-coupling term $A_\mathcal{O}[T_\gamma] \cdot F_\gamma$ is therefore not a gauge-invariant scalar but the frame-relative contribution that absorbs the path-ordering dependence; Cor. 1 packages $\eta_\mathcal{O}$ and this term into $\rho_\mathcal{O}$, whose norm bound is frame-independent up to C_{frame} by Lem. 2. $\square$ $\square$

Corollary 1 (Observer-relative FFTC residue). *Define the FFTC residue along γ (in the trivialization Φ of Thm. 2) by*

$$\rho_\mathcal{O}(f, \gamma)(s) := \eta_\mathcal{O}(\gamma(s)) + A_\mathcal{O}(\gamma(s))[T_\gamma(s)] \cdot F_\gamma(s).$$

(Distinct from the curvature 2-form $\kappa_\mathcal{O} = dA_\mathcal{O} + A_\mathcal{O} \wedge A_\mathcal{O}$; $\rho_\mathcal{O}$ is a section along γ , $\kappa_\mathcal{O}$ is an $\text{End}(E)$ -valued 2-form on $|\mathcal{K}_t|$.) Then Eq. (8) reads $(D_\mathcal{O}F_\gamma)(\gamma(s))[T_\gamma(s)] = f(\gamma(s)) + \rho_\mathcal{O}(f, \gamma)(s)$, and

$$\|\rho_\mathcal{O}(f, \gamma)(s)\|_{h_\mathcal{O}} \leq \varepsilon_\mathcal{O} + \|A_\mathcal{O}\|_{g_\mathcal{O} \otimes h_\mathcal{O}, \infty} \cdot \|F_\gamma(s)\|_{h_\mathcal{O}}.$$

The right-hand side is frame-independent up to C_{frame}^2 by Lem. 2. Moreover, $\rho_\mathcal{O}(f, \gamma) \equiv 0$ for every $\mathcal{O}$ -regular f and every γ in the classical limit $s_\mathcal{O} \rightarrow 0$ (equivalently, $K_\mathcal{O} \rightarrow \delta$ and $A_\mathcal{O} \equiv 0$), recovering the classical FTC $(dF_\gamma/ds)/\|\dot{\gamma}\|_g = f(\gamma(s))$.

Remark 9 (Scaling of the residue bound with curve length). The bound $\|\rho_\mathcal{O}(s)\|_{h_\mathcal{O}} \leq \varepsilon_\mathcal{O} + \|A_\mathcal{O}\|_\infty \cdot \|F_\gamma(s)\|_{h_\mathcal{O}}$ grows through the $\|F_\gamma\|_{h_\mathcal{O}}$ factor, which by Step 4 is $O(L(\gamma|_{[a,s]}))$. The residue is therefore “ $\mathcal{O}$ -small” in the regime $\varepsilon_\mathcal{O} \ll 1$ and $\|A_\mathcal{O}\|_\infty \cdot C_{\text{frame}} \cdot (\|f\|_\infty + \varepsilon_\mathcal{O}) \cdot L(\gamma|_{[a,s]}) \ll 1$; i.e., for short or moderately-long curves relative to the inverse connection norm. The Hypothesis Surface pipeline (main paper, §5) operates in this regime by construction: per-stage displacement budgets are bounded and anti-masking keeps $\varepsilon_\mathcal{O}$ small.

Remark 10 (Relation to Observer-Relative Stokes). Theorem 2 and Thm. 1 share the connection $\nabla_\mathcal{O}$ but extract different scalars from $\kappa_\mathcal{O} = dA_\mathcal{O} + A_\mathcal{O} \wedge A_\mathcal{O}$. The FFTC is a pointwise along- γ statement: $D_\mathcal{O}$ applied to F_γ and contracted with $T_\gamma(s)$. The Stokes theorem is a loop-integrated statement: $D_\mathcal{O}f$ integrated around an $\mathcal{O}$ -closed boundary equals the curvature flux plus an interaction residue plus a boundary holonomy. We do not identify the pointwise connection term $A_\mathcal{O}(\gamma(s))[T_\gamma(s)]F_\gamma(s)$ with the surface integral $\iint_\Sigma \kappa_\mathcal{O} \cdot f$; such an identification would require a homotopy hypothesis we do not post. The Stokes theorem is used for the main paper’s loop-level integrability arguments, not for Thm. 2.

5. Integrability Corollary and Scope

Corollary 2 ($\mu = 0$ bounds the FFTC residue). The anti-masking rate μ (Invariant 2 of the main text) tracks the fraction of hypotheses emitted without bounded-resolution fibers. “Bare” hypotheses correspond to kernel responses with effectively unbounded support, which inflate $A_\mathcal{O} \wedge A_\mathcal{O}$ in Def. 1

and also break $\mathcal{O}$ -regularity (Def. 6). Enforcing $\mu = 0$ restores both bounds: $\|A_{\mathcal{O}}\|_{g_{\mathcal{O}} \otimes h_{\mathcal{O}}, \infty} < \infty$ and $\|\eta_{\mathcal{O}}\|_{h_{\mathcal{O}}} \leq \varepsilon_{\mathcal{O}}$, which by Cor. 1 yields

$$\|\rho_{\mathcal{O}}(f, \gamma)(s)\|_{h_{\mathcal{O}}} \leq \varepsilon_{\mathcal{O}} + \|A_{\mathcal{O}}\|_{g_{\mathcal{O}} \otimes h_{\mathcal{O}}, \infty} \cdot \|F_{\gamma}(s)\|_{h_{\mathcal{O}}}$$

throughout $[a, b]$. The empirical section of the main paper verifies $\mu \approx 0$ across five domains.

Remark 11 (Scope of the $\mu \rightarrow \rho_{\mathcal{O}}$ bridge). *The mapping from the text-level diagnostic μ to bundle-level bounds in Cor. 2 is declarative, not quantitative: we assert that $\mu = 0$ restores the hypotheses under which Thm. 2 and Cor. 1 hold, not that there is an explicit function g with $\|\rho_{\mathcal{O}}\| \leq g(\mu)$. Constructing such a g (for example, via a concrete embedding of the pipeline’s claim/fiber emission into $(E, h_{\mathcal{O}}, \nabla_{\mathcal{O}})$) is deferred as adjacent work and is flagged as a limitation of the main paper. A rigorous quantitative construction for one such embedding (flat $\mathbb{R}^n$ base, trivial rank- N bundle, Gaussian-bump fibers with the natural a_i -weighted connection) is carried out in the companion supplement Anti-Masking Bridge: A Quantitative $g(\mu)$ for One Embedding, yielding $\|\rho_{\mathcal{O}}\| \leq G_0 + G_1\mu + G_2\mu^2$ with coefficients explicit in $(N, s_{\mathcal{O}}, R_{\min}, \varepsilon_{\text{bare}}, C_{\text{frame}}, L)$.*

Remark 12 (Pipeline-emission of (H3)). *Hypothesis (H3) (piecewise Lipschitz at scale $s_{\mathcal{O}}$) is independent of $\mathcal{O}$ -regularity (H2) and is posted separately in Rmk. 13. In the Hypothesis Surface pipeline of the main paper, emission of f bounds $\|\nabla \sigma_i\|$ per fiber at the emission stage, which implies (H3) with $L_f \leq \max_i \|\nabla \sigma_i\|_{\infty}$. This is an instantiation claim about one implementation, not a theorem about the proof’s hypothesis list; readers using a different pipeline should verify (H3) directly.*

Remark 13 (Scope and standing hypotheses). *This document proves Part 1 only, in its reduced form: Eq. (8) for the kernel-only line integral of Def. 5, and the Stokes identity Eq. (3) for simply connected Σ with $\mathcal{O}$ -closed boundary. Standing hypotheses:*

(H1) $A_{\mathcal{O}}$ is $\mathcal{O}$ -bounded: $\|A_{\mathcal{O}}\|_{g_{\mathcal{O}} \otimes h_{\mathcal{O}}, \infty} < \infty$.

(H2) $f \in \Gamma(E)$ is $\mathcal{O}$ -regular (Def. 6).

(H3) f is piecewise Lipschitz at scale $s_{\mathcal{O}}$: there exists $L_f < \infty$ such that $\|f(y) - f(z)\|_{h_{\mathcal{O}}} \leq L_f \cdot \|y - z\|_{g_{\mathcal{O}}}$ whenever $\|y - z\|_{g_{\mathcal{O}}} \leq s_{\mathcal{O}}$ and y, z lie on a common smooth segment of f . Independent of (H2).

(H4) Kernel-scale hypothesis: $s_{\mathcal{O}} < \text{inj}(M, g)$.

(H5) (Local to Thm. 1 only:) the surface Σ is simply connected with non-empty boundary, and $\partial\Sigma$ is $\mathcal{O}$ -closed (Def. 7). Not posted globally.

Deferred. Part 2 (integral of a derivative yields a boundary term plus a symbolic holonomy $\mathcal{H}_{\mathcal{O}}$ in a dual sense) and the full symbolic-integration operator with the memory/distortion term E_{acc} (Rmk. 4). The reduced Part 1 proved here is the load-bearing result for the Hypothesis Surface paper’s claim that bounded observation leaves an irreducible residue $\rho_{\mathcal{O}}$, and that integration under $\mu = 0$ converges with bounded residue norm.